

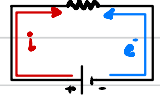
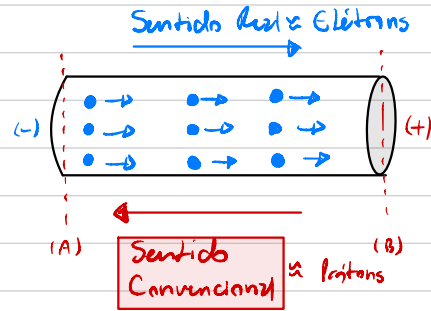
ELETRODINÂMICA

Fluxo

1 Corrente Elétrica (i)

↳ Fluxo ordenado de cargas elétricas

* Só há formação de **corrente** se houver **dif** entre (A) e (B).

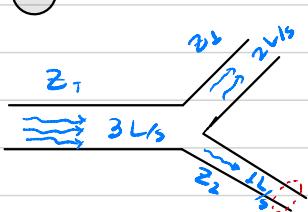


Corrente $\approx$ Fluxo $\approx$ Vazão

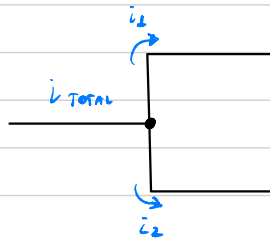
$$i = \frac{\text{Carga (Q)}}{\text{tempo (\Delta t)}}$$

$$Z = \frac{V_{ab}}{\Delta t}$$

I

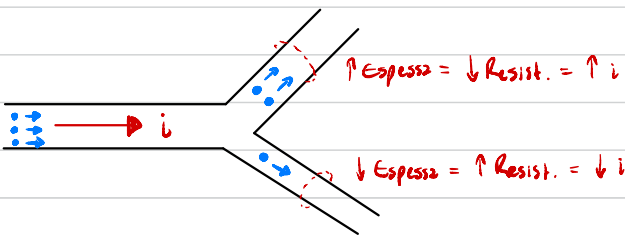


$$Z_{\text{TOTAL}} = Z_1 + Z_2$$



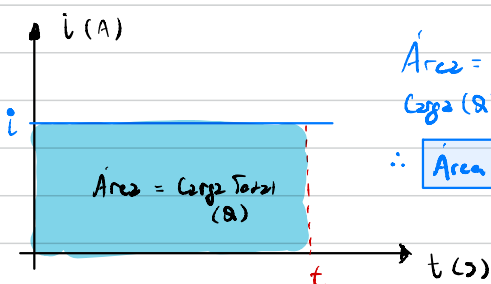
$$i_{\text{TOTAL}} = i_1 + i_2$$

II



1.1 Gráfico (i x t)

$$i = \frac{Q}{\Delta t} \Rightarrow Q = i \cdot \Delta t$$



$$\begin{aligned} \text{Área} &= b \cdot h \\ \text{Carga (Q)} &= i \cdot t \\ \therefore \text{Área} &\approx Q \end{aligned}$$

• Unid. de Medida

$$[i] = \frac{[Q]}{[t]} = \frac{1 \text{ Coulomb}}{1 \text{ segundo}} = 1 \text{ Ampère}$$

* Baterias eletrônicas:

$$[mAh] = \text{Carga (Q)}$$

$$i = \frac{Q}{t}$$

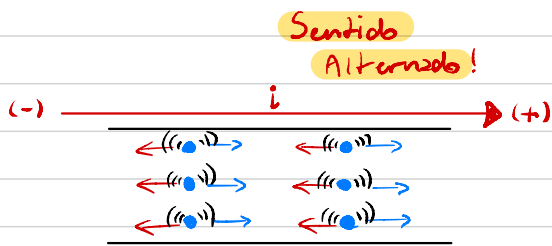
$$\text{Ex: } 4000 \text{ mAh} = 4 \text{ A} \cdot \text{h}$$

P) Carregar essa bateria é necessário $i = 4 \text{ A}$ por 1 h .

$$4 \text{ A} \rightarrow 1 \text{ h} \quad 4 \cdot 1 = 2 \cdot x$$

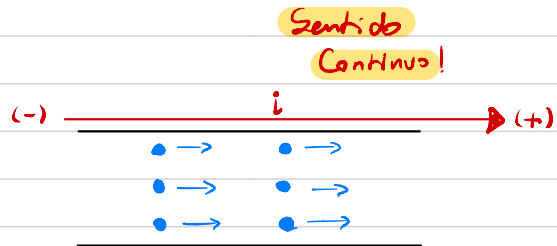
$$2 \text{ A} \rightarrow x \quad x = 2 \text{ h}$$

①.2 Corrente Alternada vs Corrente Contínua



Indução Magnética
(Lei de Faraday-Lenz)

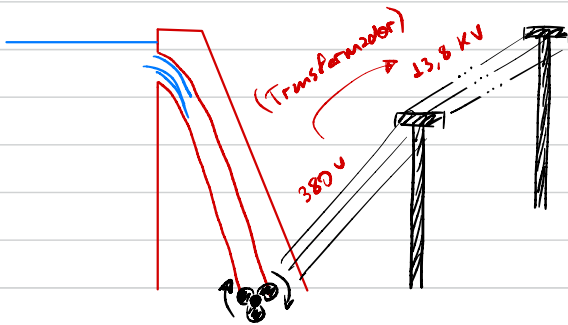
VARIAÇÃO de FLUXO MAG
Gera Corrente Elétrica



"Sensível"

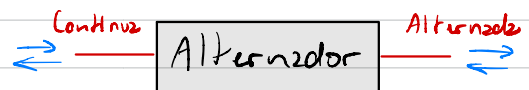
Aplicações:

- 1) Geração/Transmissão de Energia
- 2) Transformadores
- 3) Instalações Elétricas
- 4) Ventiladores, Motores



Aplicações:

- 1) Circuitos Eletrônicos
- 2) Pilhas e Baterias
- 3) Placas Fotovoltaicas
- 4) Circuitos de carros/motos



• Guerre das Correntes:

↳ Nikola Tesla vs Thomas Edison

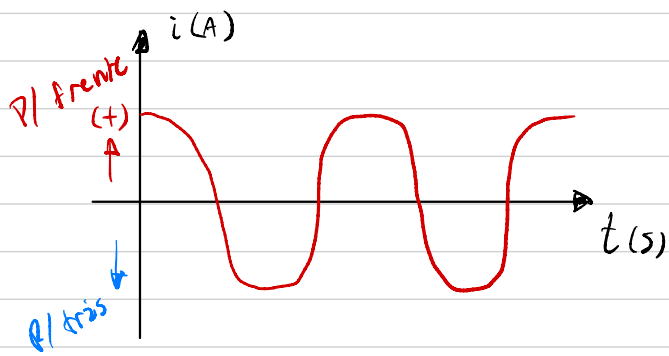


Corrente
Alternada

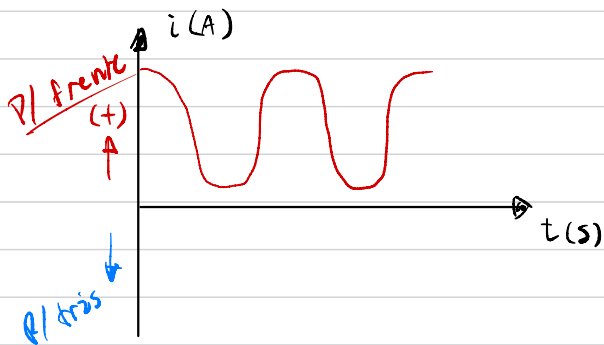
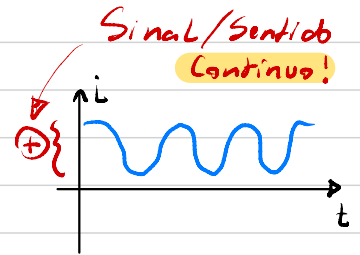


Corrente
Contínua

• Gráfico Corrente Alternada vs Continua



Corrente Alternada!



Corrente Contínua!